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Fig. 3: Wireless fence transmitter and collar

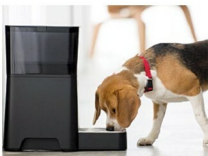


Fig. 4: Petnet smartfeeder being used to feed a Beagle

operation. Some use the chip ID as their database index and print it on receipts, test results, vaccination certifications and other records.

The microchip

A microchip implant is a passive RFID device. It remains inactive until it is powered by the scanner as it lacks an internal power source. The microchip contains three elements: a chip (an IC), a coil inductor (possibly with a ferrite core) and a capacitor.

The chip contains unique identification data and electronic circuits to encode the information. The coil acts as the secondary winding of a transformer, receiving power inductively coupled to it from the scanner. The coil and capacitor together form a resonant LC circuit tuned to the frequency of the scanner's oscillating magnetic field to produce power for the chip. The chip then transmits data back through the coil to the scanner. These components are encased in a biocompatible soda-lime or borosilicate glass, and hermetically sealed. Barring rare complications, dogs and cats are unaffected by these.

In dogs, the chip is usually insert-



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Fig. 5: Petnet smartfeeder iPhone application

ed under the skin at the back of the neck, between the shoulder blades on the dorsal midline. It can often be felt under the skin. Thin layers of connective tissues form around the implant and hold it in place.

Implanted microchips can distort magnetic resonance imaging (MRI), including those of the spinal cord.

Some disadvantages of microchipping are:

1. Chips could move and cause a variety of problems.
2. Different brands of microchips need different scanners.
3. All places do not have the correct scanners needed or any scanners at all.
4. Foreign body put under the skin has a risk of causing tumours.

Universal scanners have been introduced that detect microchips resonating at multiple frequencies.

According to Dr Gautam Unny (B.V.Sc, M.I.S.V.S), microchipping has huge potential. Only one per cent of dogs are microchipped in India, whereas all can be. Also, every shelter and clinic requires a scanner.